JUNHOO LEE

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EDUCATION

Seoul National University Seoul, South Korea
Ph.D. Candidate in Intelligence and Information Sep 2021 – Present (Exp. Aug 2026)
Advisor: Prof. Nojun Kwak, Machine Intelligence & Pattern Analysis Laboratory (MIPAL)

Seoul National University Seoul, South Korea
B.Sc. in Electrical and Computer Engineering Mar 2017 – Sep 2021
B.S. Thesis: *Improvement of Elevator Control Algorithm Using Reinforcement Learning.* Advisor: Sungroh Yoon

RESEARCH INTERESTS

My research studies how foundation models acquire reusable structure, and how that structure can be controlled, adapted, and diagnosed across language, vision, generative modeling, and embodied interfaces. My work on discrete diffusion language models introduces inference-time planning structure for controllable generation [C1], while my meta-learning papers study fast adaptation and task generalization through optimization-trajectory structure and flexible episodic interfaces [C6, C7]. I also work on model diagnosis, semantic robustness, and data-efficient transfer, including deep support vectors for pretrained-model interpretation [C5], black-box semantic fingerprinting for generative models [C2], multimodal temporal grounding [C4], and instruction-following failures in text-to-image generation [W1]. Recently, I have been extending this line toward vision-language-action and embodied foundation models, where pretrained semantic structure must remain reliable under changing instructions, visual observations, and action interfaces.

PUBLICATIONS

[C1] **Unlocking the Potential of Diffusion Language Models through Template Infilling**
Junhoo Lee, *Seungyeon Kim, Nojun Kwak*
*Annual Meeting of the Association for Computational Linguistics (**ACL**)*, 2026.
Oral Presentation (485/10518=4.61%)

[C2] **CSF: Black-box Fingerprinting via Compositional Semantics for Text-to-Image Models**
Junhoo Lee, *Mijin Koo, Nojun Kwak*
*The IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**)*, 2026.

[C3] **Deep Edge Filter** (Co-first)
*Dongkwan Lee†, **Junhoo Lee**†, Nojun Kwak*
*The Conference on Neural Information Processing Systems (**NeurIPS**)*, 2025.

[C4] **What's Making That Sound Right Now? Video-centric Audio-Visual Localization**
Hahyeon Choi, **Junhoo Lee***, Nojun Kwak*
*The IEEE/CVF International Conference on Computer Vision (**ICCV**)*, 2025.

[C5] **Deep Support Vectors**
Junhoo Lee, *Hyunho Lee, Kyomin Hwang, Nojun Kwak*
*The Conference on Neural Information Processing Systems (**NeurIPS**)*, 2024.

[C6] **Any-Way Meta Learning**
Junhoo Lee, *Yearim Kim, Hyunho Lee, Nojun Kwak*
*The AAAI Conference on Artificial Intelligence (**AAAI**)*, 2024.

[C7] **SHOT: Suppressing the Hessian along the Optimization Trajectory**
Junhoo Lee, *Jayeon Yoo, Nojun Kwak*
*The Conference on Neural Information Processing Systems (**NeurIPS**)*, 2023.

[W1] **Do Not Think About Pink Elephant!** (Co-first)
*Kyomin Hwang†, Suyoung Kim†, **Junhoo Lee**†, Nojun Kwak*
CVPR Workshops – Responsible Generative AI, 2024.

[W2] **Coreset Selection for Object Detection**
Hojun Lee, Suyoung Kim, **Junhoo Lee***, Jaeyoung Yoo, Nojun Kwak*
CVPR Workshops – Dataset Distillation, 2024.

[W3] **Practical Dataset Distillation Based on Deep Support Vectors**
Hyunho Lee, **Junhoo Lee***, Nojun Kwak*
ECCV Workshops – Dataset Distillation Challenge, 2024.

[J1] **End-to-End Multi-Entity Customization**
Wonhark Park, Jaehyun Lee*, Wonsik Shin,* **Junhoo Lee***, Nojun Kwak*
IET Image Processing, Accepted. 2026.

[J2] **The Role of Teacher Calibration in Knowledge Distillation**
Suyoung Kim, Seonguk Park, **Junhoo Lee***, Nojun Kwak*
IEEE Access, 2025.

AWARDS & HONORS

ICML Gold Reviewer	2026
BK21 Future Innovation Talent Bronze Prize (USD 1,000)	2023
BK21 Outstanding Research Talent Fellowship (USD 3,500)	2023
Yulchon AI Star Scholarship (USD 8,000)	2022
3rd Place, SNU FastMRI Challenge (USD 3,000)	2021
Kwanak Scholarship (full funded)	2021
National Science and Engineering Scholarship (full funded)	2017–2018

RESEARCH PROJECTS

Gradient-Descent-Based Meta-Learning Algorithms (Samsung Electronics, Main Researcher, 2022–2023)
Self-Directed Visual Object Detection for Shopping Assistance (IITP, Project Manager, 2022–2024)
AI Video Generation for Korean Traditional Media Art (MCST/EASYWITH, Main Engineer, 2024–2026)
Physically Plausible Video Diffusion Modeling (AI Star Fellowship/MSIT, Project Manager, 2025).

SKILLS

Programming Languages: Python, CUDA, R, C++, Verilog; **ML Development Stack**: PyTorch, TensorFlow, Slurm, Docker.

TALKS

Compositional Semantic Fingerprinting (CSF) for Black-box Attribution of Text-to-Image Models, SWCS 2026, Apr. 2026
Deep Support Vectors: Interpreting Deep Models via Support Vector Extraction, KCC 2024, Jun. 26, 2024.

ACADEMIC SERVICE

Gold Reviewer: ICML 2026.
Conference Reviewer: CVPR (2025, 2026), ICCV (2025), ECCV (2026), ICLR (2025, 2026), ICML (2026), NeurIPS (2024, 2025, 2026).

PATENTS

Method and Apparatus for Edge Filtering to Improve the Robustness of Deep Learning Models, Korean Patent 10-2025-0215458
Method and Apparatus for Controlling Unintended Object Generation in Image Generation Models through Prompt Reconfiguration, Korean Patent 10-2025-0063788.